

Jane Rose Ferdinant

Subject Matter Expert – Data & Statistical Chart Analysis (AI Benchmark)

M.Tech (Software Systems, CUSAT) • Assistant Professor @ SNMIMT • Ex-Technovalley Technical Consultant
Remote (India / Global) • Email: janeferdinant@gmail.com • GitHub: github.com/Janeferdinant

Application Dossier
AI Benchmark Standards
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INTERACTIVE DEPLOYMENT URLS
Direct verified links to the live interactive web portfolio and downloadable PDF documents:

🌐 Live Web Portfolio | 📄 Deployed AI Benchmark PDF | 📋 Academic & Industry CV | 💻 GitHub Repo

18+	95% CI	0.94	p < 0.001
Frontier Models Audited	**Bootstrap Protocol**	**Inter-Rater Agreement**	**Significance Criterion**
MMLU-Pro, GSM8K, SWE-bench	B = 10,000 Resamples	Cohen's Kappa (κ) Standard	FDR / Bonferroni Corrected

EXECUTIVE PROFILE & BACKGROUND

Academic educator, data scientist, and AI benchmarking SME with an M.Tech in Information Technology (Software Systems) from **CUSAT**. Currently serving as Assistant Professor at **SNMIMT** teaching Deep Learning, AI, and Soft Computing; previously Technical Consultant at **Technovalley Software India** and Assistant Professor at Sri Vasavi Engineering College. Published IEEE researcher in computer vision. Brings rigorous mathematical methodologies, non-parametric bootstrapping, hypothesis testing, contamination forensics, and visual chart translation to AI benchmark governance.

SPECIALIZED AI BENCHMARK EVALUATION PROJECTS

1. Multi-Model AI Benchmark Analytics Engine & Pareto Frontier

Statistical Frontier Modeling

Evaluates frontier models (Claude 3.5 Sonnet, GPT-4o, DeepSeek-R1, DeepSeek-V3, Llama 3.3 70B, Qwen 2.5 72B) across MMLU-Pro, GSM8K, SWE-bench Lite, and HumanEval, mapping multi-objective Pareto efficiency curves across Accuracy vs. Latency (ms) and Inference Cost ($/1M tokens).

- **Confidence Bounds:** Employs 10,000 non-parametric bootstrap iterations to compute exact 95% Confidence Intervals, preventing false claims of superiority when scores fall within the error band.
- **Tradeoff Discovery:** Proved that while DeepSeek-R1 leads in complex STEM reasoning (84.0% on MMLU-Pro), Claude 3.5 Sonnet (79.2%) is 2.7x faster (420ms vs 1,150ms), defining the optimal low-latency Pareto frontier for production SLAs.

Model	Family	MMLU-Pro	95% Confidence Band	Std Error (SE)	Latency (ms)	Pareto Efficient?
DeepSeek-R1	Reasoning-RL	**84.0%**	[82.6%, 85.3%]	±0.68%	1150 ms	**Yes ★**
Claude 3.5 Sonnet	Proprietary	**79.2%**	[77.8%, 80.5%]	±0.69%	420 ms	**Yes ★**
GPT-4o	Proprietary	**77.4%**	[75.9%, 78.8%]	±0.74%	380 ms	**Yes ★**
DeepSeek-V3	Open-Weights	**75.9%**	[74.4%, 77.3%]	±0.74%	310 ms	**Yes ★**
Llama 3.3 70B	Open-Weights	**71.8%**	[70.2%, 73.3%]	±0.79%	290 ms	**Yes ★**

2. Benchmark Anomaly & Contamination Forensic Suite

Data Integrity & Outlier Audit

Engineered an automated data integrity auditor detecting memorization vs. generalization through token perplexity shifts and multi-domain Simpson's Paradox audits.

- **Contamination Flagging ($Z > 2.5$):** Measured $\Delta \text{PPL} = \text{PPL}_{\text{unseen}} - \text{PPL}_{\text{test}}$. Discovered that fine-tuned open models claiming >90% GSM8K collapsed to 48.6% when numerical operands were perturbed, demonstrating test-set memorization.
- **Simpson's Paradox Resolution:** Disaggregated macro benchmark scores into stratified sub-domains (Law, Medicine, Math, CS). Showed that Model Beta led macro scores (+1.4%) solely through high-volume CS syntax while suffering severe regressions (-12.4%) in Legal reasoning.

3. Bradley-Terry Elo Matrix & LLM-as-a-Judge Calibration

Tournament Modeling & Bias Mitigation

Constructed a head-to-head pairwise tournament model across 14,500 blind battle matches, resolving systematic evaluator distortions.

- **Length & Position Bias Elimination:** Identified uncalibrated judges granting an unmerited +7.3% win rate boost to verbose answers regardless of substance. Implemented a ridge regression length penalty, improving alignment with human annotators to $r = 0.91$ and Cohen's $\kappa = 0.89$.
- **Radar Capability Profiling:** Multi-dimensional mapping of models across Reasoning, Math, Coding, Instruction Following, Safety, and Hallucination Resistance.

FOUNDATIONAL APPLIED AI & DATA SCIENCE PROJECTS

Harvestable Black Pepper Recognition (CV / ResNet50): Custom ResNet50 model detecting and classifying ripeness of black pepper in agricultural fields. Published in *IEEE Xplore / ICACC 2019*.

Stock Market Forecasting (LSTM & Deep RL): Financial time-series price trend prediction, risk volatility modeling, and algorithmic trading strategy evaluation.

COVID-19 & Pneumonia Automated Detection (CNN): Deep transfer learning model for rapid radiographic detection from digital chest X-rays.

E-Commerce Collaborative Filtering Recommendation: Matrix factorization and item-user recommendation engine boosting discovery and conversion metrics.

APPLIED INDUSTRY CASE STUDIES

Case 01: Uncovering Simpson's Paradox in Multilingual Reasoning Benchmarks

Problem: Candidate model claimed a +4.1% aggregate gain across 14 languages, but regional users suffered sentiment degradation.
Resolution: Discovered 20x higher English sampling masked an 8.4% drop in Indic/African languages. Enforced stratified population weighting.

Case 02: Forensic Audit of Math Test-Set Leakage in 70B Models

Problem: Open-weights 70B model claimed higher GSM8K performance than models 5x its size.
Resolution: Accuracy dropped from 91.2% to 48.3% on perturbed operands, confirming rote memorization. Delivered forensic report for leadership.

Case 03: Eliminating Position & Length Biases in Automated LLM Judges

Problem: LLM judges favored first-presented and excessively verbose answers.
Resolution: Formulated Order A/B swapping and an orthogonal length penalty, raising human agreement from $\kappa = 0.62$ to $\kappa = 0.89$.

METHODOLOGICAL COMPETENCY & TECHNICAL STACK

Statistical Methods

- Non-parametric Bootstrapping (10k)
- Paired Wilcoxon & Welch's t-test
- Multiple Testing (Bonferroni / FDR)
- Bradley-Terry Tournament Modeling
- Cohen's Kappa (κ) & Gwet's AC1

AI Evaluation Frameworks

- MMLU-Pro, GSM8K, MATH
- SWE-bench (Verified & Lite), HumanEval
- HELM, MT-Bench, LMSYS Arena
- TruthfulQA & Hallucination Auditing
- Inference Latency (TTFT) & TPS Analysis

Data & Engineering Stack

- Python (SciPy, Statsmodels, Pandas, NumPy)
- SQL, PostgreSQL, BigQuery, MATLAB
- Chart.js, D3.js, Matplotlib, Seaborn, Power BI
- PyTest, Git, CI/CD, Vercel, Docker
- Executive & Technical Communication